Φάση Δ: Big Data approach με Spark

Υπεύθυνος: Data Engineer (bonus)

Pipeline: Imputer → StringIndexer → OneHotEncoder → VectorAssembler → Scaler → Classifier

Βήματα:

1. Εγκατάσταση PySpark
2. Φόρτωση CSV σε Spark DataFrame
3. Δημιουργία ML Pipeline
4. Εκπαίδευση & αξιολόγηση

```python
import json
from pyspark.sql import SparkSession
from pyspark.sql.types import DoubleType
from pyspark.sql.functions import col, when, lit
from pyspark.ml import Pipeline
from pyspark.ml.feature import StringIndexer, OneHotEncoder, VectorAssembler,
StandardScaler, Imputer
from pyspark.ml.classification import LogisticRegression

spark = SparkSession.builder \
    .appName("Stroke_Pipeline") \
    .master("local[*]") \
    .getOrCreate()

df = spark.read.option("header", True).option("inferSchema", True) \
    .csv("../healthcare-dataset-stroke-data.csv").drop("id")

# Φόρτωση feature metadata για αναφορά
with open("../data/feature_metadata.json", "r", encoding="utf-8") as f:
    feature_meta = json.load(f)

print(f"Raw CSV: {df.count()} rows, {len(df.columns)} cols")
print(f"Feature metadata: {feature_meta['feature_dim']} διαστάσεις")
```